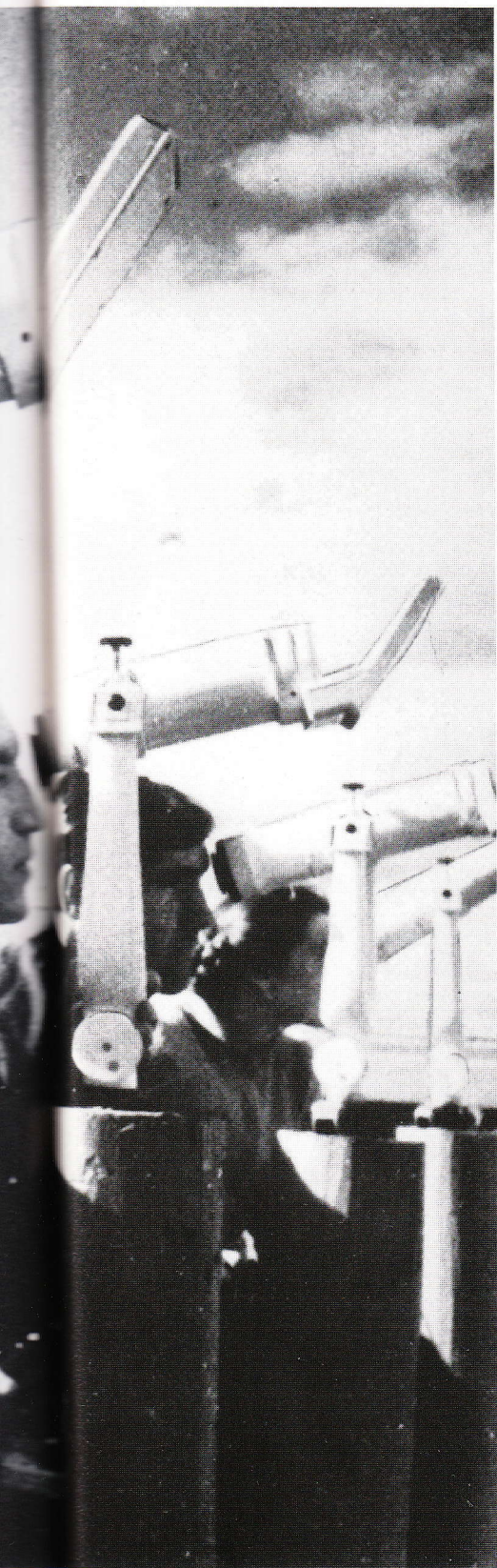
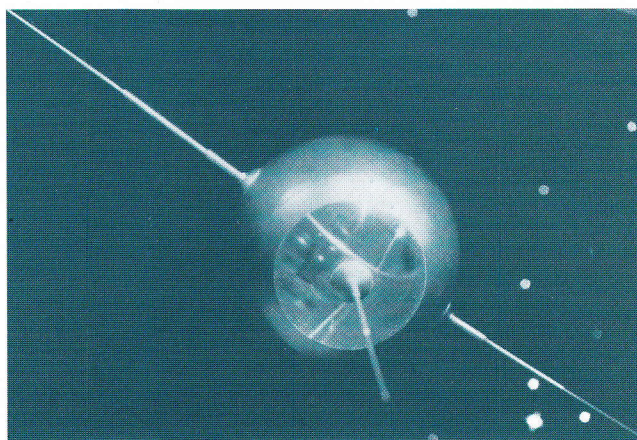




FORTY YEARS AGO THIS MONTH
the Soviet Union orbited a
“man-made moon” whose
derisive chirp persuaded
Americans they’d already lost
a race that had barely begun

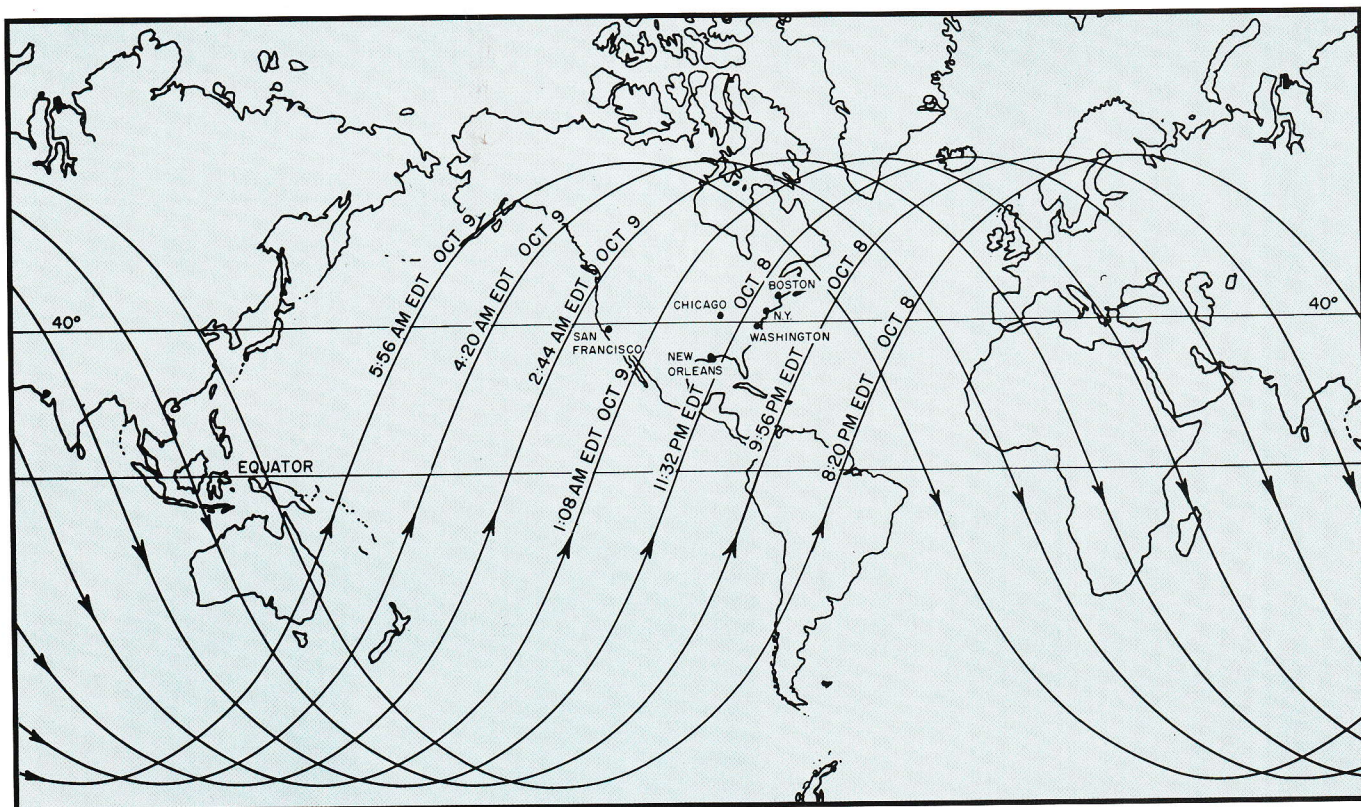
SPUTNIK



BY EDWIN DIAMOND AND STEPHEN BATES

IT WASN'T THE BEST OF TIMES, BUT IT WASN'T THE worst of times either. Although a mild recession had cooled down the post-Korean War economy, many families were living comfortable lives in the autumn of 1957. There were 170 million Americans now, and more of them had taken a vacation that summer than ever before, just like the swells out in Southampton.

To be sure, there was turbulence in the air. Three years after *Brown v. Board of Education* had struck down school segregation, the governor of Arkansas, Orval Faubus, defied a federal judge's integration order. Reluctantly, President Dwight D. Eisenhower dispatched the 101st Airborne to enforce the Constitution at Little Rock's Central High School. Slowly, though, the walls of segregation were falling. In July of that long-ago summer, Althea Gibson of Harlem, U.S.A., scored a first. A decade after Jackie Robinson had broken the baseball color line, Gibson won the Wimbledon tennis singles championship



Four days after the launch the U.S. Naval Research Laboratory released a chart of *Sputnik's* course for October 8 and 9.

and curtsied to the Queen of England.

No Americans were fighting abroad in 1957, though tens of thousands of GIs were deployed in Cold War hot spots from divided Berlin to the Korean demilitarized zone. The Americans and the Russians were methodically testing bigger and "dirtier" (more radioactive fallout) nuclear bombs while perfecting intercontinental missile systems to deliver them. But Nikita Khrushchev, the new Number One Red (as the newspapers referred to him), was talking peaceful competition between socialism and the Free World (as the same papers referred to our side), and summitry, not shooting, seemed to be the prospect between the United States and the U.S.S.R.

Everything in fact appeared to be converging on a broad consensual middle, a prospect that evoked varying responses. What enthusiasts touted as serene abundance (the Republicans had just produced a film called *These Peaceful and Prosperous Years*), critics

scorned as soulless conformity and complacency—from Holden Caulfield's contempt for "phonies" to *The Man in the Gray Flannel Suit's* self-doubt.

Then, early in the evening of October 4, the sky seemed to fall, literally, on the American edifice. At 6:30 P.M. EST the Associated Press moved a bulletin: Moscow Radio had announced that "the Soviet Union has launched an earth satellite." Later in the evening NBC interrupted regular programming to give more details of the "man-made moon" and to play its high-pitched radio signal "as recorded by RCA engineers." The next morning's *New York Times* and *Washington Post* both gave three-line eight-column banners to the feat, the kind of headline reserved for a Pearl Harbor or a D-day. The editors of *Newsweek* scrapped their planned feature on Detroit's new line of cars (trashing 1,309,990 cover copies—twenty tons of paper). The new cover showed an artist's conception of the Soviet satellite *Sputnik* (Russian for "fellow traveler"). Inside, the weekly explained "The Red Conquest," "The Meaning to the World," and, ominously, "Why We Are Lagging."

OVERNIGHT THE self-assured center began coming apart. Inventive, free-enterprise America, home of Edison and the Wright brothers, Levittown and "modern laborsaving kitchen appliances," was being overtaken—surpassed?—by a backward, totalitarian, *Communist* nation. And the shock to can-do pride was the least of it. A missile gap apparently yawned, with the Soviets pulling decisively ahead in the ultimate nuclear weapons, ICBMs. Democrats in Congress charged that amiable Ike's mid-register budgetary caution had jeopardized U.S. military prowess. It seemed that the energetic five-star architect of victory in the Big War had turned into a Burning Tree Country Club slacker (one cartoonist showed *Sputnik* whizzing past a golf ball), a myopic Pangloss, a President Magoo.

Sen. Stuart Symington of Missouri, a potential Democratic presidential candidate in 1960, demanded that the President call a special session of Congress to address the *Sputnik* crisis. Ike refused; he even declined to deliver

a televised speech addressing the nation's apprehensions, so as not to appear "alarmist." Instead he chose to hold a news conference on October 9, five days after the Soviet announcement. It was, in the view of Eisenhower's biographer Stephen Ambrose, one of the most hostile Q&A sessions of Ike's Presidency. Eisenhower repeatedly maintained that *Sputnik* was in essence meaningless. "As far as the satellite itself is concerned," he said, "that does not raise my apprehensions, not one iota."

The reporters were, to put it mildly, skeptical. If the Soviets could orbit a satellite, they could fire a nuclear warhead across the ocean at Washington;

Eisenhower fields questions from a hostile press at his October 9 news conference. Below, the British press response.



Sputnik, said Ike, "does not raise my apprehensions, not one iota." The reporters were, to put it mildly, skeptical.

if the Americans couldn't orbit a satellite, they couldn't shoot a warhead inside Soviet boundaries. Or so experts were telling journalists, and journalists were telling the public. War planners had been confident that the United States could fight off a nuclear strike launched by Soviet long-range bombers, but if Soviet *missiles* could reach the United States, as *Sputnik* hinted, then perhaps our way of life *was* doomed. *Life* magazine presented "The Case for Being Panicky."

Readers were persuaded. Soon Gallup found that half of all Americans believed the Soviets held the lead "in the development of missiles and long-distance rockets." By early 1958 more than a third of Americans thought that the Soviets "could





American dogs picket the U.N. as Laika orbits in *Sputnik II*; below, one of Laika's colleagues in his space suit.

wipe out most cities in the United States in a matter of a few hours with their new rockets and missiles." One out of three Americans also expected the outbreak of World War III by the early 1960s.

The news grew gloomier. In early November, just in time for the celebration of the fortieth anniversary of the Russian Revolution, the Soviets announced the launch of a second *Sputnik*, this one carrying Laika, the orbiting dog. No matter that Laika had a one-way ticket; the capsule accommodated enough oxygen to keep her alive for ten days (because a thermal-control system failed, she didn't last even that long). The Soviets began hinting at plans for manned spaceflights. "What will Americans find on the other side of the moon?" went the joke. "Russians."

In reality the United States was winning both the arms race and the nascent space race. There was indeed a "missile gap," but the lead belonged to the Americans. Ike, it turned out, knew something the rest of us didn't.

First dogs, and soon, the Soviets hinted, men. "What will Americans find on the other side of the moon?" went the joke. "Russians."



Sputnik, like so many 1950s developments in military technology, had its roots in World War II. Hours after the satellite was launched, a U.S. military official complained, "We've got the wrong Germans!" He was mistaken. Most of "their" German rocket scientists—the Peenemünde group captured by the Soviets at the end of the war—had been repatriated. The four-hundred-thousand-pound three-stage rocket that launched *Sputnik* may have been an elaboration of the successful German V-2 design, but it was homegrown.

The best known of "our" Germans, those who had fled west to avoid capture by the Russians, was a civilian scientist for the U.S. Army Redstone missile command in Huntsville, Alabama, named Wernher von Braun. Smooth, handsome in a Hollywood-heavy sort of way, von Braun had been trying for years to get the government to make satellites a priority. His 1954 report "A Minimum Satellite Vehicle" outlined a plan for orbiting an American satellite by 1956. "It is only logical to assume that other countries could

do the same," von Braun wrote, adding (emphasis in original): "*It would be a blow to U.S. Prestige if we did not do it first.*" He sought one hundred thousand dollars to start an Army satellite program. The request was turned down.

Instead the administration divvied up the tasks—and the pork—of missile development. In that 1950s spirit of compromise, every service got a piece of the action (as did the contractors allied with each service—Martin, Northrop, Convair, Aero-Jet General). Even though von Braun and the Army were far ahead in testing rocket designs, the Defense Department gave the Navy the satellite assignment. A new Navy Vanguard rocket would be developed to lift a four-pound satellite and its modest telemetry into orbit. At the same time, the Army would continue to develop the Jupiter intermediate-range ballistic missile, while the Air Force would work on its Atlas and Titan intercontinental ballistic missiles. Eisenhower didn't warm to the idea of a "missile czar," on the model of Gen. Leslie Groves during the Manhattan Project, to knock Army, Navy, and Air Force heads together; each service forged ahead independently of the others.

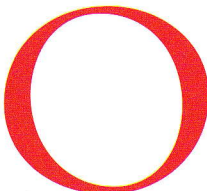
On July 31, 1955, American scientists, with the blessing of the White House, announced that the United States would launch a satellite during the International Geophysical Year (IGY), a resolutely peaceful eighteen-month (July 1957 through the end of 1958) multinational investigation of the planet and its resources. Within days of the American announcement, Soviet scientists revealed that they, too, were readying a satellite, also to be launched during IGY. American experts scoffed.

While the Navy worked to meet the IGY goal, von Braun's Army team launched a four-stage rocket on September 20, 1956. It reached a speed of 13,000 miles per hour and a record-setting altitude of 682 miles. The last stage might have been capable of achieving orbit, but because the Navy was in charge of satellites, the nose cone was filled with sand.

To Eisenhower and one faction of his allies, *Sputnik* was noise signify-

ing nothing. The President said, "We never considered ourselves to be in a race." The White House adviser Sherman Adams declared that the United States had no interest in "an outer-space basketball game." In *The New Republic* Richard Strout dryly saw a parallel to Ike's above-the-fray re-election campaign of 1956: "Mr. Eisenhower appeared prepared to treat the satellite as though it were Adlai Stevenson."

But some politicians took to bad-mouthing one another. A few Republicans, including Ike's Vice President, Richard M. Nixon, blamed Harry S. Truman, by then nearly five years out of office. Truman responded by writing a long article blaming Eisenhower and lamenting this "sorry chapter in the story of our defense." Distributed by the North American Newspaper Alliance, the article made its way to the front page of *The New York Times*. Evoking the memory of the "atomic spy couple," Julius and Ethel Rosenberg, *U.S. News & World Report* suggested that skulduggery explained the Soviet feat: "Did Russia Steal Satellite Secret from U.S.?"

 OTHERS FAULTED progressive educators, who, the critics claimed, had concentrated on children's feelings to the detriment of hard knowledge. *Life* devoted five issues to the "Crisis in Education," arguing in one article that the spartan Soviet system was producing students better equipped to cope with the Space Age. *Why Johnny Can't Read* rocketed up the bestseller lists. In an October 31 news conference, Eisenhower, while remaining unruffled by *Sputnik*'s military implications, declared himself "shocked" to learn the magnitude of the nation's education shortcomings. The following year he backed the National Defense Education Act, which funded laboratories and textbooks in public schools as well as loans for college-bound students—the federal government's first major steps into education. Like interstate highways, schooling had become

a matter of national defense, endorsed by both the Republican White House and the Democratic-controlled Congress. The former Harvard president James Bryant Conant urged parents to tell children, "For your own sake and for the sake of the nation, do your homework."

Still others contended that the Soviets' lead in space reflected a deteriorating American spirit. Peace and prosperity, according to this line of argument, had produced an indolent self-satisfaction. "Our goal has become a life of amiable sloth," wrote the journalist Thomas Griffith. To the sharp-tongued playwright and Republican loyalist Clare Boothe Luce, *Sputnik*'s beeps represented "an intercontinental outer-space raspberry" aimed at American pretensions of superiority. *Sputnik*, then, might serve as a warning shot that would force "bland, gray-suited" America to contemplate national interest instead of self-interest. "We needed *Sputnik*," Adlai Stevenson said, calling the satellite "sure proof that God has not despaired of us."

Vice President Nixon recognized that the White House efforts to shrug off *Sputnik* were failing miserably. He deemed Adams's basketball remark "wrong in substance and disastrous in terms of public opinion." In a San Francisco speech the Vice President staked out his own ground, saying that "we could make no greater mistake than to brush off this event as a scientific stunt of more significance to the man in the moon than to men on earth." Privately Nixon urged Eisenhower to say that money was no object in the contest of freedom against slavery. Ignoring the advice, Ike instead cautioned against any "hasty and extraordinary effort under the impetus of sudden fear."

Nixon wasn't the only Republican up in arms. In early November a panel of defense experts delivered its previously commissioned report to Ike's National Security Council. The work of such Establishment bulwarks as John McCloy and Paul Nitze, the report argued that the United States could be "critically vulnerable" to a



As television recorded the scene, Vanguard's first-stage rocket roared, spewed flame and smoke—and rose all of four feet.



America's first satellite goes to ground on December 6; the fiasco drew taunts from the press (and the Russians).

missile attack by the end of 1959, with likely casualties of up to 50 percent. Even if the Soviets chose not to wage cataclysmic war, the report suggested, they could conquer space, maybe militarize the moon. The panel's bottom line: Continued American security required *major* increases in the military budget, to be achieved by deficit spending. Called the Gaither Report (after the group's chair, the Ford Foundation's head, H. Rowan Gaither, Jr.), the document was leaked to *The New York Times* and

the *Washington Post*. In his magisterial memoir *Danger and Survival*, McGeorge Bundy writes that Ike came to feel he'd been hit by a "double barrelled shock"—*Sputnik* and Gaither.

But the worst blast came a few weeks later. Along with portraying the *Sputniks* as meaningless, the administration had been telling reporters that the United States was about to launch a satellite of its own. The Navy was still in charge of the program, though von Braun and the Army were quietly at work too. The day after *Sputnik I's* launch, von Braun had told Defense Secretary Neil McElroy that the Navy rocket "will never make it," whereas the Army rocketeers could launch a satellite in sixty days. McElroy took it under advisement. A month later, amid the aftershocks of *Sputnik II*, von Braun was told to get to work.

The Navy's Vanguard program was also on an accelerated countdown. The original schedule called for a dozen meticulous test runs, each one involving additional hardware and equipment. Not until number seven was a full-scale "earnest try" for orbit to be attempted. But with two *Sputniks*

in orbit, as *Newsweek* put it, "the Vanguard test rocket with its grapefruit-sized satellite suddenly became the U.S. answer to the Soviet challenge."

FOR FOUR EXCRUCIATING days beginning on December 2, the formerly sleepy (and off-limits) Cape Canaveral test site on Florida's Atlantic coast became media central. The Vanguard launch team sweated through a series of postponements that Navy spokesmen attributed to, at one time or another, "balky" guidance systems, "minor electrical troubles," and "sticky LOX" (liquid oxygen) valves. One countdown was aborted because of the "weariness of overworked technicians." Because this was nominally IGY "science," reporters received extraordinary cooperation: two Air Force flat-bed trailers for photographers, schedules of launch times, viewing points on the beach outside the Cape gates.

Finally, at 11:45 A.M. on December 6, a rocket propulsion engineer flipped the firing toggle. As television recorded the scene, Vanguard's first-stage rocket roared, spewed flame and smoke, rose four feet—and fell back onto the steel launching pad and tumbled to the ground, exploding in a spectacular fireball. The satellite cargo, thrown clear, was damaged but still beeping. For Eisenhower, recovering from a late November stroke, the news couldn't get much worse. Headline writers around the world outdid each other: "Flopnik," "Stay-putnik," "Dudnik." A Russian delegate to the United Nations asked his American colleagues if they would be interested in applying for aid "under the Soviet program of technical assistance to backward nations."

Ike, his popularity plummeting (it had gone from almost 80 percent in late 1956 to just 50 percent in late 1957), bent a bit. His post-*Sputnik* budget increased military expenditures, a rise that mandated, in Eisenhower's words, "at least a token reduction in the 'butter' side of government," so spending on urban development and hospitals was cut. Still, the parsimoni-

ous President had little use for space exploration. "Look," he told his cabinet, "I'd like to know what's on the other side of the moon, but I won't pay to find out this year." Even so, he signed legislation creating NASA in 1958.

THROUGH IT ALL, though, while Ike bent, he never broke. He remained mostly, and bafflingly, unflappable in the eyes of many Americans. He had his reasons. He knew that *Sputnik* actually represented some good news. While it showed that the Soviets were ahead in rocketry thrust, it also showed that they were well behind in miniaturizing communications technology (the beeping *Sputnik I* weighed 184 pounds). Moreover, *Sputnik* settled practically a lingering question of international law: How high does a nation's airspace reach? That issue would have considerable bearing on the spy satellites then under development, a space venture that *did* interest Eisenhower. "The Russians," Assistant Defense Secretary Donald Quarles told the President, "have in fact done us a good turn, unintentionally, in establishing the concept of freedom of international space."

Most important, Ike knew that the Russians were behind the United States militarily—so far behind on warhead production and ICBM development that a surprise attack on the United States would be suicidal. Since 1956 Ike had been seeing photographs taken by supersecret U-2 spy planes. These photos revealed the Soviet disadvantage in ICBMs and tellingly, they didn't show any preparations for a first strike. "We can still destroy Russia," Eisenhower told his cabinet. "We know it."

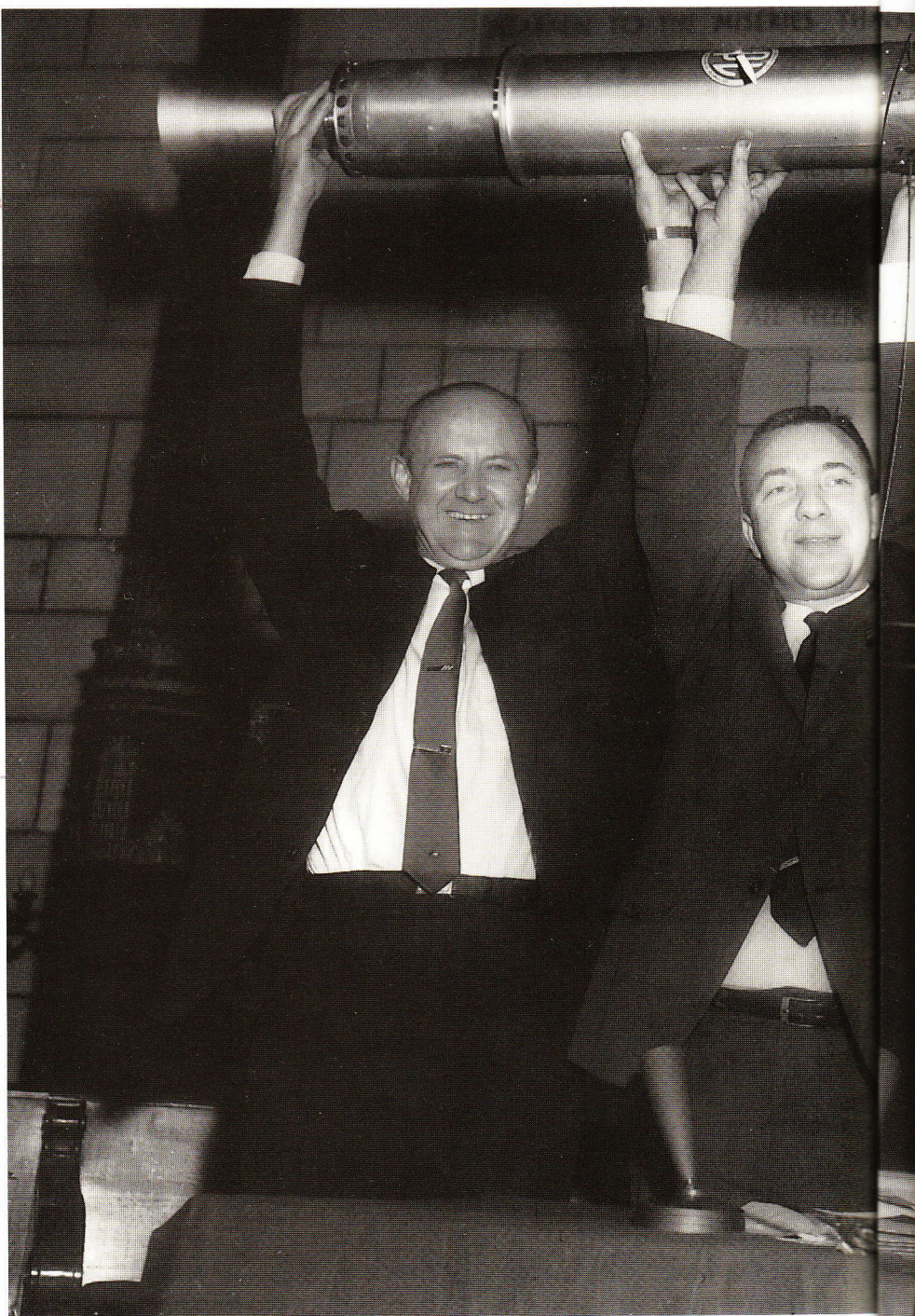
But the U-2 information was top secret. The Gaither Report authors didn't know about it. Neither did a thirty-four-year-old Harvard professor named Henry Kissinger, author of a sky-is-falling report for the Rockefeller Brothers Fund that was heavily publicized in early 1958, as brother Nelson positioned himself for 1960 presidential politics (when Nelson appeared on

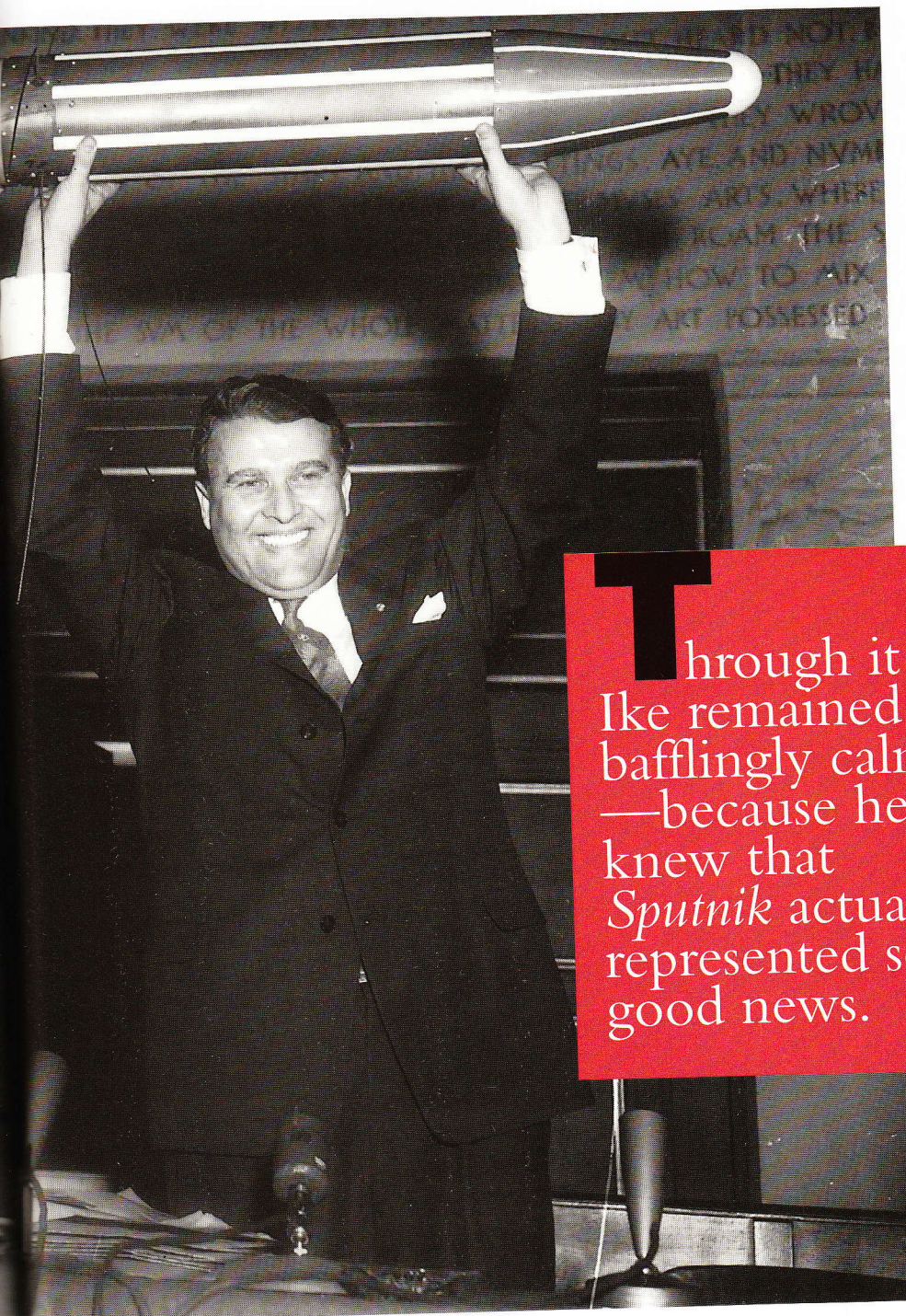
Scientists loft a model of the first U.S. satellite, *Explorer*, celebrating its successful launch in February of 1958. Dr. Wernher von Braun is at right.

"Today" on NBC, Dave Garroway offered to send a free copy of the report to anyone who requested it. More than two hundred thousand people asked). Ike couldn't reveal the U-2 flights without disclosing American violations of Soviet airspace; at the altitudes at which the U-2s were flying, international law was very clear. This

could jeopardize his attempts to achieve accommodation with Khrushchev over such flash points as Berlin and nuclear testing. (It later turned out that the Soviets already knew of the U-2 overflights but didn't have the missiles to shoot them down—yet.) Ike also didn't want to push the Soviets to escalate their military spending, as any revelation of American superiority would likely do. So mostly he tried to persuade the public to trust him on the basis of his own military record.

Historians tend to give Eisenhower





Through it all Ike remained bafflingly calm—because he knew that *Sputnik* actually represented some good news.

high marks for *Sputnik*. In Stephen Ambrose's view, Ike's calm response to the Soviet satellite was "one of his finest hours," saving his country countless billions of dollars. In his book *Grand Expectations*, James Patterson agrees, noting that Ike presided over major gains in the U.S. nuclear capacity and did so quietly enough to allay Soviet fears.

Historians, though, do fault Ike for failing to grasp the public relations implications of *Sputnik*. The National Security Council recommended a

greater emphasis on space-related projects "which, while having scientific or military value, are designed to achieve a favorable worldwide psychological impact," but Eisenhower responded coolly. As he later said, "I don't believe in spectaculars." When von Braun's Army team successfully launched the first American satellite (which weighed just thirty-one pounds) on January 31, 1958, Eisenhower downplayed what others were portraying as a great American triumph. Ike instructed his aides, "Let's not make

too great a hullabaloo over this."

History's verdict on Ike came slowly. Even after the U-2 flights became known, the Alsop brothers, columnists Joseph and Stewart, argued that because the photography was limited, it was not all that trustworthy. U-2 spy photography, they said, was confined to major Soviet railroad lines that would service ICBM launch complexes and thus skipped large parts of the Soviet land expanse. Later, by the time John Kennedy was in the Oval Office—propelled to some extent by all the talk of the space and missile gaps—U.S. spy satellites covered all of the Soviet Union. These photos, in the summer of 1961, confirmed that there was indeed a missile gap all along—in America's favor. Mac Bundy, a key player in the Kennedy White House, recalls the Soviet missile threat being steadily de-escalated from hundreds of ICBMs during the late 1950s to around thirty-five by the mid-1960s. The Alsops eventually confessed their error.

Such was the self-defeating effect of Soviet secrecy: In the absence of facts about Russian ICBMs, many Americans responded out of fear. By the late 1960s the American nuclear triad ensured invulnerability, reliability, and massive retaliatory capacity. By the 1990s the Communist state had collapsed, amid evidence that the Soviets had spent themselves into poverty trying to keep up in all the various races. ★

*Edwin Diamond, who died in July, taught at New York University. His book *White House to Your House: Media and Politics in Virtual America* (M.I.T. Press) was issued in May in a paperback edition with a new epilogue. Stephen Bates is the literary editor of the *Wilson Quarterly*. Their article on the ancient history of the Internet ran in the October 1995 issue.*